FOUNDATIONS OF GENERATIVE AI (ICT4403)

Assignment 2: Autoregressive Text Generation System

Submission Document & GitHub Repository Verification Sheet

1. Required Submission Details (Step 3)

Student Name	Saksham Gupta
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Assignment Title	Foundations of Generative AI (ICT4403) — Assignment 2
Title of the Developed Autoregressive Text-Generation System	Comparative Word-Level Autoregressive Text Generation on Lewis Carroll's Alice's Adventures in Wonderland using Simple RNN, Bidirectional RNN, LSTM, and GRU
Name and Link of the Dataset Used	Alice's Adventures in Wonderland (Lewis Carroll, Project Gutenberg eBook #11) Dataset Link: https://www.gutenberg.org/files/11/11-0.txt
GitHub Repository Link	Repository Link: https://github.com/Saksham-Gupta-GH/fgai-autoregressive-text-generation-alice

2. GitHub Repository Structure & Verified Deliverables (Step 2)

All implementation code, pre-trained model checkpoints, interactive notebooks, plots, and experimental data are hosted in the public GitHub repository above in strict compliance with the assignment guidelines:

- Complete Source Implementation (**`src/`**): Modular PyTorch implementation containing `preprocessing.py` (chronological 80/10/10 split, train-only vocabulary fitting with `min_freq=2`), `models.py` (Simple RNN, BiRNN with $[h_n[-2], h_n[-1]]$ state concatenation, LSTM, and GRU), `train.py` (training loop with validation checkpointing), `evaluate.py` (loss, perplexity, Top-1/Top-5 accuracy), and `generate.py` (temperature sampling, Distinct-1/2, trigram repetition metrics).
- Interactive Jupyter Notebook (**`notebooks/text_generation_assignment2.ipynb`**): Complete, fully executed notebook with all 11 cells containing data preparation, model training progress, evaluation plots, and generated text outputs.
- Automated End-to-End Pipeline (**`run_experiments.py`**): Single-command reproduction script executing the full experimental workflow: preprocessing, training, evaluation, plotting, and generation.
- Empirical Results & Visualizations (**`results/`**): Tabular CSV reports, generated sample logs across temperatures ($T \in \{0.2, 0.5, 1.0, 1.5\}$), and high-resolution comparison plots (`loss_curves.png`, `perplexity_comparison.png`, `accuracy_comparison.png`, etc.).
- Comprehensive Technical Report (**`README.md`**): Complete report covering problem formulation, dataset characteristics, architecture details, experimental protocol, full evaluation tables, temperature analysis, and model justification.

3. Summary of Key Experimental Findings

- Best Performing Model: LSTM achieved the strongest next-word prediction performance (Best Val Loss: 3.9903, Val Perplexity: 54.07, Held-Out Test Top-1 Accuracy: 23.29%, Test Top-5 Accuracy: 47.23%), with GRU performing closely behind (Val Loss: 4.0353, Test Top-1: 22.53%) with 11% fewer parameters.
- Sampling Temperature Optimization: Temperature $T = 0.5$ demonstrated the optimal empirical trade-off between lexical diversity (Distinct-1: 0.60–0.73) and structural coherence, eliminating repetition loops while maintaining fluent narrative dialogue tags.

Submitted in fulfillment of Steps 3 and 4 of the Assignment 2 submission procedure for ICT4403.